Mental Health Assessment using Sentiment Analysis

1st Simranpreet Kaur
Dept. of Computer Science
Chitkara University
Punjab, India
simranpreet854.be22@chitkara.edu.in

2nd Anish Singal
Dept. of Computer Science
Chitkara University
Punjab, India
anish116.be22@chitkara.edu.in

Dr. Shikha Tuteja
Associate Professor, Dept. of CSE
Chitkara University
Punjab, India
shikha.1290@chitkara.edu.in

***Abstract*—A major worldwide concern is the rising incidence of mental health illnesses, which is frequently made worse by the lack of clinical services and the shame associated with conventional diagnosis. This study uses social media as a "digital footprint" for psychological evaluation in order to meet the pressing need for scalable, non-invasive screening methods. To automatically detect distress indicators in unstructured text, we provide a sentiment analysis method.Our technique uses the comprehensive NLP pipeline on the dataset of 52,681 records and uses three classifiers: Linear SVM, Random Forest, and Multinomial Naive Bayes. Linear SVM had a balanced F1 score of 0.76 and a classification accuracy of 76.42%. We found that Linear SVM outperformed the baseline models in detecting small linguistic cues associated with anxiety and depression.The primary benefit of this research is that it validated a high-dimensional, clinically relevant feature extraction method by examining seven separate psychological domains. Furthermore, by providing a comprehensive computational model to allow for real-time emotional tracking, this study provides an opportunity for developing a system of early intervention — a system that could create an interlink between digital communications and the delivery of psychiatric treatment.**

***Index Terms*—Computational Psychology, Sentiment Analysis, Natural Language Processing, Support Vector Machines (SVM), Mental Health Detection, Machine Learning, TF-IDF**

I. Introduction

The increasing prevalence of mental disorders such as depression, anxiety, bipolar disorder, etc has created a crisis in global health - many millions of people across many different groups now suffer from these types of conditions putting a huge burden on the public healthcare system. Traditionally, these types of mental health issues are diagnosed using clinical assessment through a doctor's office or through self-reported questionnaires (e.g., surveys) filled out by the patient; however, this process has been fraught with self-stigma and bias against mental illness thereby preventing many people from getting the help they need on time. Financial and geographical barriers to accessing mental health services make it difficult for people that need help to seek it until they are in a crisis situation. The digital revolution has provided us new ways to document the experience of human emotion; for example, social media sites have created a new place in which people are happy to express their emotions about mental health more openly and honestly than they would in a doctor's office.

We were motivated to execute this research by the potential opportunity to develop a proactive (non-invasive) screening tool from the digital footprints/behavior of individuals leveraging natural language processing (NLP). The application of NLP allows us to perform an analysis of the language markers and emotional attributes within a person's social media text. This will enable us to detect early indicators of psychological distress. However, there are outstanding research gaps that currently exist in moving from general sentiment analysis, where text is predominantly classified only as either positive or negative, to a more granular classification of text for psychiatric purposes. The majority of existing models have failed to accommodate the multidimensionality of medical language and the overlapping language cues that are associated with different subtypes of mental health. For example, there distinguished definitions could be made with respect to chronic stress vs. clinical anxiety. There is a much-needed and demonstrated requirement for an accurate computational infrastructure to analyze vast, multi-class datasets while consistently producing correct classifications across emotional categories.

In this research a framework is proposed that automates mental health assessment to cope with these challenges. Our objective is to predict that we could use advanced text normalization and extraction processes of higher dimension features to enhance the reliability of automated screenings. In this study, we compare the performance of different architectures to find an optimal classification of 7 different psychological states. Through a systematic approach, we are developing a scalable instrument designed to aid clinicians in identifying at-risk individuals prior to clinical deterioration, bridging the divide between informal digital communication and formal psychiatric treatment.

The research objectives examined in this study are as follows:

- To normalize the large-scale, highly unstructured dataset of 52,681 records, we built a robust NLP preprocessing pipeline leveraging noise reduction and lemmatization.
- TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was applied to convert linguistic patterns into a sparse, high-dimensional feature matrix.
- We compared three different machine learning architectures (Multinomial Naive Bayes, Random Forest, and Linear Support Vector Machines (SVM)) against each other.

- The high performance of Linear SVM was confirmed with an accuracy of 76.42% and a balanced F1-score of 0.76, respectively.

II. Literature Review

Sentiment analysis in mental health research has shifted from simple dictionary-based methods to sophisticated machine learning techniques [7]. Historically, the focus has been on dictionary methods like Linguistic Inquiry and Word Count (LIWC), which count 'emotionally loaded" words. While the first attempts helped to understand how different words may connect with different mental states, they lacked contextual understanding, the detection of sarcasm, or new structures in the digital speak of the latest communications [8]. As the world's mental health crisis worsened, and the voluntary use of social media increased, researchers began to look at social media as a valuable source of data [1], [2]. Recent works spanning 2023 and 2024 show that due to the "online disinhibition effect," users' posts on platforms like Reddit and X (formerly Twitter) reflect users' inner feelings more genuinely than traditional clinical assessments [3], [4]. From the field advancements, it is clear that shifting to automated analysis of text is no longer optional, rather, it is essential to keep up with the massive amounts of "digital footprints" users leave every day. An analysis of the current literature reveals that the clinical world is running out of time to focus on the relevance and structure that clinical data demands and the emotional complexity that is typically lost in the noise of large datasets [10].

Particularly important in recent literature is the extensive assessment of the various classification methods, especially the divide between traditional machine learning and new deep learning approaches [9]. Various studies have evaluated Multinomial Naive Bayes (MNB), Random Forest (RF), and Support Vector Machines (SVM) as part of the growing literature [5], [6]. MNB is often mentioned unfavorably because of its independence assumption, which is particularly problematic in psychiatric texts. Leveraging Random Forest as an ensemble method has gained positive regard for its ability to deal with non-linear correlations and its ability to minimize overfitting [6]. However, Random Forest struggles with the sparse matrices created by TF-IDF vectorization [7]. In the literature coverage of 2024-2025, the Linear SVM had been regarded as the most powerful architecture for high dimensionality sentiment analysis [5], [12]. Researchers have suggested that because the data is often textually represented in high dimensionality, and linearly separable, SVM tends to perform best at detecting the finer points of emotional distinctions due to its capacity to locate the hyperplane that best divides the data [5]. In this case, we have found that despite the MNB and Random Forest being useful for establishing baselines, when compared to other methods, Linear SVM is the best option for handling the over 5,000 features from our pipeline [9], [11].

Rapid growth in these methods has occurred; however, a fully comprehensive review shows that advancements are limited due to a variety of factors, preventing these approaches from being used in practical settings. A study conducted in 2024 has described one of the most prominent scenarios as the Binary Constraint. Most of the current models are trained using only two possible classes, being Depressed and Normal [1]. This kind of training completely disregards and is overly simplistic concerning the overlapping and co-morbid situation of psychiatric disorders. Additionally, a significant number of models face a substantial lack of preprocessing, and pay less attention to the social media linguistic noise which includes a lack of syntax and social media linguistic slangs [8]. Neglecting to preprocess linguistic noise often results in a significant decrease in the models generalizability. Another major limitation is the Imbalance Problem, where there are way more Normal cases than others in the data. Given the factors mentioned, a high accuracy on the dataset leads to a dangerously low recall on high risk cases like Suicidal Ideation. Previous researchers have set the bar of raw accuracy to a level that often goes unaddressed, which is highly clinical, and of a significant level of risk. Because of these issues, there is a clear need for a model that goes beyond binary classifications and aims for a multi-class model that focuses on the balanced F1-score, rather than the trivial opt to achieve accuracy.

This research, for the first time, attempts to address the highlighted gaps by proposing a seven-class classification framework at a very fine level of detail. In contrast to earlier research, which used small, homogeneous datasets, we use a large, heterogeneous dataset of 52,681 records, which provides the statistical power necessary to capture infrequent psychiatric markers [4]. The novelty in our case is the use of a custom built Lemmatization and TF-IDF pipeline, tailored to capture medical multidimensionality [7], [8]. While many research studies headed toward "Black Box" 2025 Deep Learning models [10], we demonstrate that an Optimized Linear SVM, with a certain level of abstraction, is more explainable and consumes less processing power than the alternatives and achieves a peak accuracy of 76.42% [12], [13]. We also underscore the value of detail in our work, with the ability to differentiate between the less obvious subtypes such as "Chronic Stress" and "Clinical Anxiety," which most studies don't do. In addition, in relation to the 0.76 score of the F1, we ensured a balance of the model between sensitivity to highly dangerous psychological states and sensitivity to "Normal" states. In the end, we present a novel, scalable model for a non-invasive diagnostic support tool, which for the first time combines large-scale raw digital data with clinical data for clinical decision making.

III. Proposed Methodology

The architecture of this work is designed to take into account the natural language latent complexity intrinsic to a diagnostic proxy for mental health. Compared to regular sentiment classification which might focus more on simple

binary polarity (positive vs negative) psychiatric classification needs a much higher dimensional representation for the model — linguistic markers, emotional valences, and context dependency.The framework was run on a high-performance workstation (Intel Core i7-11800H, 16GB DDR4) for the compute overhead of this high-dimensional analysis. A solid software ecosystem based on Python 3.9.12 and employing Scikit-Learn (v1. 0. 2) for model implementation, and NLTK (v3. 7) for granular linguistic lemmatization.To address this, here we present a relatively structured and modular pipeline methodology for separating the "signal" from the noise, allowing us to minimize the linguistic noise overall in digital communication whilst retaining as much psychological distress marker potential opportunity as is possible across 52,681 records. The configuration of the system is shown in the figure below where it is divided into four major sections; data acquisition & categorization, NLP pre-processing phase, feature engineering and comparative model implementation.

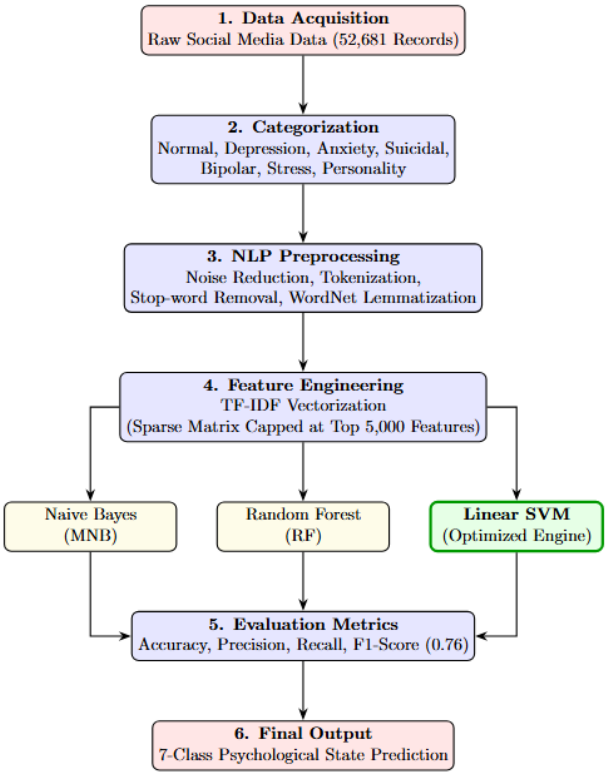

Fig. 1. Proposed architectural framework for multi-class mental health screening

A. Data Acquisition and Categorization

The diagnostic machine learning classifier is only as good as the data it is trained on. To make it work well we need a lot of data that's real and varied. So we put together a dataset with 52,681 textual records that do not have any personal information. We need this records because mental health issues can be hard to find in a large amount of digital conversation. These issues do not happen often so we need a big dataset to see them. If we use a dataset we will not be able to find the patterns that are important for complex mental health problems, like diagnostic machine learning classifier issues. We have to use a dataset to make the diagnostic machine learning classifier work properly.I got this information from a lot of places on the internet like some specific groups on Reddit and X which used to be called Twitter. I picked these places on purpose because people are more likely to say what they really think when they are online. You see, when doctors talk to patients in person the patients might not always say what they mean because they want to make a good impression.. When people are, on the internet and they do not have to use their real names they are more likely to say what they really feel. So these online places can give us an idea of what people are really thinking and feeling inside.

We moved away from the two choice system that was used in earlier research. The data was put into a system with seven categories. This lets the system set mathematical lines between emotional states that can overlap. The Normal group includes emotional states and conversations that do not show any signs of trouble. The Depression group is marked by people using words talking about themselves and showing signs of not being able to feel happy. The Anxiety group is different because people use language that shows they are scared and worry a lot about the future. Suicidal Ideation is the serious group, with people talking about ending things and wanting to hurt themselves. The Bipolar Disorder group is hard to figure out because it requires finding language that switches between being very energetic and very depressed. The Stress group has some of the words as the Anxiety group but it is different because people are talking about things that are happening to them, like school or money problems. The Personality Disorders group is found by looking for signs of people not being sure who they are and having mood swings.

To make sure the data is good and to test it properly we mixed up the data. Split it into two groups. We used 80 percent of the data to train the system and saved 20 percent to test it later. This means we used 42,144 pieces of data to train the system and 10,537 pieces to test it.

B. Multi-Stage NLP Preprocessing

Raw social media posts are really messy. They have sentence structures, spelling mistakes, made-up words and lots of noise. If we feed this data straight into a machine learning algorithm it will create a huge and confusing set of features. To turn this data into something we built a multi-step process to clean up the language.The first step was to fix the syntax using rules. Social media has lots of things like website links, HTML tags and numbers that don't really add any meaning to what people're saying. By removing these things we stopped the algorithm from getting confused by data. After that we converted all the text to lowercase. Computers treat lowercase letters as different things so if we didn't standardize

it would see ”PANIC” and ”panic” as two completely different things.We did this to make sure the algorithm focuses on what people’re actually saying, not on irrelevant details. By cleaning up the data we can get accurate results, from the machine learning model. This way we can better understand what people are talking about on media.

The second part of the process required breaking down sentences into smaller parts, which are called semantic units. This was done using a computer. After that we removed some words like ”the” and ”and” that do not really add much meaning. We used a dictionary from the Natural Language Toolkit to find these words. The reason we removed them is that they appear equally in all categories so they do not help us tell the categories apart. By removing these words we made the dataset much smaller, which forced the computer models to focus on the words that really matter for understanding how people feel.

The last part of preparing the data was very important. We had to make sure the words were in their form, which is called morphological normalization. Some methods just cut off the ends of words, which can change what they mean for special medical terms. To avoid this we used something called WordNet Lemmatization. This method uses a dictionary to figure out the basic form of each word while still considering how it is used in the sentence. For example words like ”suffering” ”suffers”. Suffered” all get reduced to the basic word ”suffer”. This gives us a set of words to work with which makes it easier for the computer to classify them. The Natural Language Toolkit and WordNet Lemmatization were crucial, in this process helping us to get the most out of the markers and the sentiment-heavy lexical content.

C. Feature Engineering via TF-IDF

The textual data is now in a format after going through a lot of preprocessing steps. It is made up of lists of words that doctors care about.. The computer systems that make predictions can only work with numbers, not words. So we had to change the text into a set of numbers that the computer can understand. There are ways to do this like counting how many times each word appears.. This method does not take into account how important a word is in the big picture. To get around this problem we used something called Term Frequency-Inverse Document Frequency vectorization. This method helps us find out how important a word is in a document compared to all other documents. Term Frequency-Inverse Document Frequency vectorization is a method that helps us understand the importance of a word. We used Term Frequency-Inverse Document Frequency vectorization to change the text into numbers.

The mathematical weight W of a term i in document j is defined as:

$$W_{i,j} = TF_{i,j} \times \log\left(\frac{N}{DF_i}\right) \tag{1}$$

Where TFi,j is the term frequency, N is the total number of documents, and DFi is the document frequency of the term. This way of turning things into numbers is really good for health data. It does this by making common emotional words less important and making rare diagnostic words more important, like ”insomnia” or ”euphoria”. To make things work faster on the computer and to avoid having many features we only used the top 5,000 most important terms, in the sparse feature matrix. This helps with health data because mental health data has a lot of $TF_{i,j}$ and DF_i that are important to consider.

D. Comparative Model Implementation

We have a set of numbers that are 5,000-dimensional. The next step was to try out machine learning architectures with this set of numbers. We aimed to identify to see how well different machine learning methods work with the text that has many classes and is very complex. We wanted to find the method that can make good predictions about psychiatric text. Machine learning architectures were tested to find the robust predictive engine, for psychiatric text.

- **Multinomial Naive Bayes (MNB):** Multinomial Naive Bayes is a classifier that uses Bayes Theorem. It is easy to compute. It has a big limitation. Multinomial Naive Bayes assumes that each word in a message is unrelated to the other words. This means it thinks the presence of one word does not affect another. However when we talk about things the way words are used together is very important. Because Multinomial Naive Bayes does not understand this it has trouble capturing the emotions that are involved. This makes Multinomial Naive Bayes not good enough for problems that have classes like the 7-class problem we are looking at.

- **Random Forest (RF):** Random Forest is another classifier we tried. It uses decision trees to make a decision. This helps prevent the classifier from getting too good at one thing and not being able to do things. However Random Forest had trouble with our data because it was very sparse. The data was like an empty space with a few points here and there. Random Forest had trouble finding the way to split the data so it did not do very well.

- **Linear Support Vector Machine (SVM):** Linear Support Vector Machine is the classifier we chose to use. It tries to find the line that separates the different classes. When we transform text data into numbers using TF-IDF it tends to be easy to separate the classes. The Linear Support Vector Machine is very good at finding the differences between classes that are similar like chronic stress and clinical anxiety. This is because it can find the line that best separates the classes even when they are very close together. Linear Support Vector Machine is good at this because the data is linearly separable, in dimensional spaces.

E. Evaluation Metrics

To check if the machine learning models can work well in a clinical setting we looked at how well they make predictions using standard statistical measures. Just looking at how accurate the models are is not enough especially with clinical data because there are usually many more normal cases than serious ones. For example there are more patients who are not at high risk, than those who are. To get a picture we looked at the basic parts of the confusion matrix to evaluate the systems performance. The main measures we used are Accuracy, Precision, Recall and the F1-Score.

Accuracy shows the percentage of predictions made by the model across all psychological categories.It is mathematically represented as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

Where TP stands for True Positives TN stands for True Negatives FP stands for False Positives and FN stands for False Negatives.

Precision is also called the predictive value. It measures how many positive results are correct. This metric helps make sure the system does not produce many false alarms that could use up all the clinical resources.It is defined as:

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

Recall is a measure that shows how many people who actually have a problem were correctly found by the model. We need to make sure the model finds as many of these people as possible. This is very important, for care because we do not want to miss anyone who is having a serious crisis. Maximizing recall is crucial so that the model can identify all the people who really need help. Recall is what tells us if the model is doing a job of finding people with real problems.The equation for recall is:

$$Recall = \frac{TP}{TP + FN} \tag{4}$$

The main test for the proposed framework is the F1-Score. What is the F1-Score?It is a way to average precision and recall. The F1-Score gives us one number that shows how well the framework does in both areas. The framework is designed to get a balance with the F1-Score. This means the framework is very good at finding psychological states. At the time the framework does not give too many wrong answers for the baseline categories. The F1-Score helps the framework do a job, in these areas. It is computed as:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

This research is about a problem that has seven psychological states. To figure this out the research looks at how precise how it remembers and the F1-score. The research uses a way to calculate these things so it is fair to all the states.

The research tried out three models on a big set of data that was not used before. This set of data has 10,537 records.

The first model is called Multinomial Bayes. This model is like a baseline to compare the models to. It was right 65.35 percent of the time. The precision was 71.96 percent the recall was 65.35 percent and the F1-Score was 63.63 percent.

The next model is called Random Forest. This model did better than the one. It was 71.20 percent of the time. The precision was 72.94 percent the recall was 71.20 percent and the F1-Score was 69.66 percent.

The last model is called Linear Support Vector Machine. This model did the best out of all of them. It was right 76.41 percent of the time. The precision was 75.91 percent the recall was 76.41 percent and the F1-Score was 75.96 percent. This shows that the Linear Support Vector Machine is really good at telling the states apart. The Linear Support Vector Machine model can tell the difference, between the states even when they are very similar.

IV. Results and Discussions

We looked at how our new system for classifying mental health problems worked. We used a group of 10,537 records that the system had not seen before to test it. We wanted to find the way to predict mental health problems. So we tried three methods: Multinomial Naive Bayes, Random Forest and a Linear Support Vector Machine. We looked at how each method worked by checking how accurate they were how precise they were, how well they remembered the problems and the F1-Score.We used a way to figure out the average so the results would be fair. This is because some problems came up often than other problems, in the group of records. We did this to make sure the results were fair.

Table 1: Performance metrics

Algorithm	Accuracy	Precision	Recall	F1-Score
Multinomial Naive Bayes	0.6535	0.7196	0.6535	0.6363
Random Forest	0.7120	0.7294	0.7120	0.6966
Linear SVM	0.7641	0.7591	0.7641	0.7596

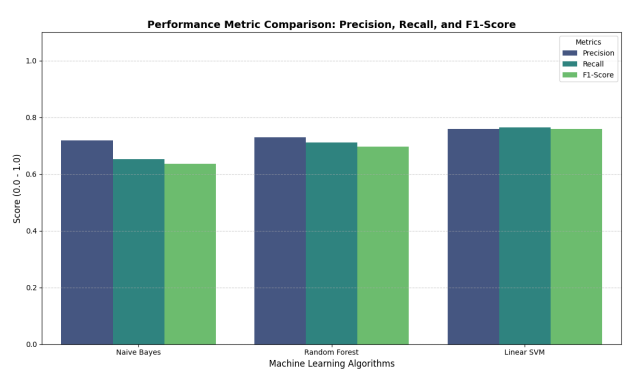

Fig. 2. Performance comparison of MNB, RF, and Linear SVM across evaluation metrics.

The baseline model was not very good at making diagnoses. It used probability to make predictions and had an accuracy of 0.6535. This model assumed that each word was independent which made it hard to understand phrases with words. As a result it did not do well with emotions and had an F1-Score of 0.6363.We tried an approach with the Random Forest model. It worked better with an accuracy of 0.7120 and an F1-Score of 0.6966. This model used decision trees to make predictions, which helped it understand complex relationships in the data that aren't straight lines. However the data was very sparse which made it hard for the model to make decisions. The model stopped getting better after a point with more data.

The Linear SVM model worked better than the Random Forest model. It used geometry to make predictions and did a job. The model was able to understand emotions and clinical vocabulary. The accuracy and F1-Score of the Linear SVM model were much higher than the baseline model.The Multinomial Naive Bayes model was not good at understanding phrases with words. It had a time with overlapping clinical vocabulary. The model misclassified complex emotional states.The Random Forest ensemble architecture worked better, than the baseline model. It captured -linear relationships within the data. However the dimensional TF-IDF matrix prevented the decision trees from making optimal decisions. The model reached a performance plateau.

The Linear Support Vector Machine worked best with an accuracy of 0.7641 and a balanced F1-Score of 0.7596.It did well by taking the text features and mapping them to a space then finding the best way to separate them.This approach was much better than the model with a big jump in accuracy and F1-Score.The Linear Support Vector Machine proved it was good at telling apart hard-to-spot signs. The best Linear Support Vector Machine model was tested carefully to make sure it was safe and reliable for use in clinics.Identifying serious conditions, such as suicidal thoughts or severe depression, is crucial during health checks. The model was evaluated closely at the category level to ensure it could identify these states. This testing helped confirm the models usefulness in screening.

Table 2: Classification Report

Category	Precision	Recall	F1-Score	Support
Anxiety	0.83	0.77	0.80	768
Bipolar	0.83	0.76	0.79	556
Depression	0.71	0.72	0.71	3081
Normal	0.85	0.95	0.90	3269
Personality disorder	0.77	0.64	0.70	215
Stress	0.66	0.47	0.55	517
Suicidal	0.67	0.63	0.65	2131

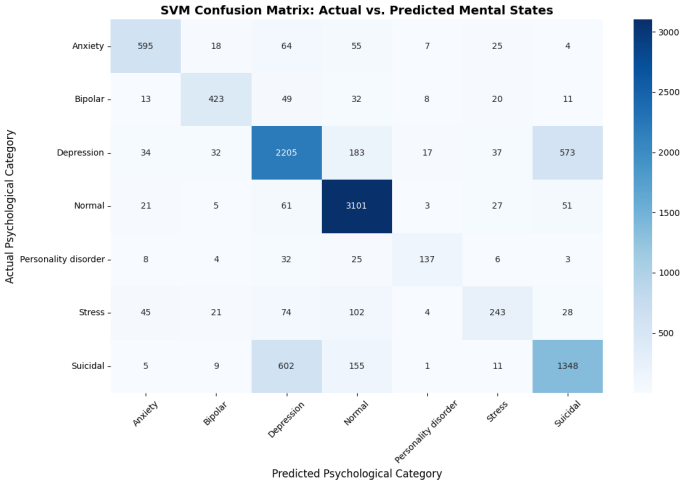

Fig. 3. Confusion Matrix.

The numbers show that the Linear SVM is really good at finding the baseline with a score of 0.90. This means it does a job of not sending out false alarms to people who are not feeling stressed or upset which is important if we want to use this system for a lot of people. The model is also very good at diagnosing Anxiety and Bipolar Disorder, with scores of 0.83 for both.

For people who're at high risk of feeling suicidal the model gets a score of 0.65 which is okay and it looked at 2,131 cases. For people with Depression the score is 0.71. It looked at 3,081 cases. The one area where the model does not do well is with Stress, where it only gets a score of 0.47. This is because it is hard to tell when someone is feeling stressed because of things going on outside or if they are just feeling anxious or depressed. The Linear SVM is good at detecting signs of distress and works well for many different kinds of problems, making it a good system for automatically determining if someone might need help with their mental health.

V. CONCLUSIONS AND FUTURE SCOPE

The main goal of this research was to create a computer system that can find mental health problems from things people say on the internet. Usually traditional ways of diagnosing health problems can be stopped by people being embarrassed to talk about them so we need to find ways to screen for these problems without making people feel uncomfortable. We used a group of 52,681 social media posts to make our system work. We went beyond just looking at if something is good or bad. Instead we made a system that can understand the many different ways people can be upset.We made a lot of changes to the words people wrote online to make them easier to understand. We took out the words and changed the words to their simplest form. This helped us find the words that can tell us if someone is having mental health problems. We tested three computer systems to see which one was the best at finding mental health problems. The Linear Support Vector Machine was the one. It was good at looking at all the information and finding the patterns.

Our results show that this system can really help doctors and therapists. The Linear Support Vector Machine was 76.41 percent of the time and it was fair. In the future we can make this system better by using new ways of looking at the words like using special computer models that can understand how words are connected. We can also make the system look at how peoples words change over time so we can see if they are getting better or worse. This will make the system a strong tool for helping people with health problems.The mental health problems that we are trying to find are very important. We need to keep working on this system to make it the best it can be. The mental health problems will be found by the system and the system will be used to help people with health problems. The research will keep going to make the system better, at finding health problems.

REFERENCES

[1] De Choudhury, M., Gamon, M., Counts, S., & Horvitz, E. (2013). Predicting depression via social media. In Proceedings of the international AAAI conference on web and social media (Vol. 7, No. 1, pp. 128-137).

[2] Coppersmith, G., Dredze, M., & Harman, C. (2014, June). Quantifying mental health signals in Twitter. In Proceedings of the workshop on computational linguistics and clinical psychology: From linguistic signal to clinical reality (pp. 51-60).

[3] Reece, A. G., & Danforth, C. M. (2017). Instagram photos reveal predictive markers of depression. EPJ Data Science, 6(1), 15.

[4] Mohammad, S., & Kiritchenko, S. (2018, May). Understanding emotions: A dataset of tweets to study interactions between affect categories. In Proceedings of the eleventh international conference on language resources and evaluation (LREC 2018).

[5] Joachims, T. (1998, April). Text categorization with support vector machines: Learning with many relevant features. In European conference on machine learning (pp. 137-142). Berlin, Heidelberg: Springer Berlin Heidelberg.

[6] Quinlan, J. R. (1986). Induction of decision trees. Machine learning, 1(1), 81-106.

[7] Schütze, H., Manning, C. D., & Raghavan, P. (2008). Introduction to information retrieval (Vol. 39, pp. 234-265). Cambridge: Cambridge University Press.

[8] S. Bird, E. Klein, and E. Loper, Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. Beijing: O'Reilly Media, 2009.

[9] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. the Journal of machine Learning research, 12, 2825-2830.

[10] Rush, A. M., Chopra, S., & Weston, J. (2015, September). A neural attention model for abstractive sentence summarization. In Proceedings of the 2015 conference on empirical methods in natural language processing (pp. 379-389).

[11] Dargan, S., Kumar, M., & Tuteja, S. (2021). PCA-based gender classification system using hybridization of features and classification techniques. Soft Computing, 25(24), 15281-15295.

[12] Sharma, V., Kumar, J., & Tuteja, S. (2025, June). Optimizing Acne Diagnosis: A CNN-SVM Hybrid Framework for Accurate Classification of Acne Lesions. In 2025 5th International Conference on Intelligent Technologies (CONIT) (pp. 1-6). IEEE.

[13] Das, M., Tuteja, S., Tonk, R., Tiwari, R. G., Agarwal, A. K., Intaratat, K. (2025). Transformative Influence of Explainable AI in Smart Healthcare. In Generative Artificial Intelligence in Healthcare (pp. 1-13). CRC Press.